

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol (max reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-max/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

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Approaches tried

- **Forward-orbit displacement (successful):** substituting $x = f(y)$ forces $f(f(y)) = 2f(y) - y$. Thus, for $d(t) = f(t) - t$, the displacement is constant along every forward orbit, whose terms are $t + nd(t)$. Positivity of all those terms gives $d(t) \geq 0$.
- **Asymptotic alignment of arithmetic orbits (successful):** for two points with positive displacements p, q , their forward orbits are arithmetic progressions. Terms of the first orbit can be aligned within a bounded distance of shifted terms of the second. The upper inequality then forces $p \leq q$; reversing the points forces $q \leq p$.
- **Connectedness after displacement quantization (successful):** once every positive displacement has one common value c , the sets where $d = 0$ and $d = c$ would both be open if both occurred. This contradicts connectedness of $\mathbb{R}_{>0}$.
- **Direct expansion of the squared inequalities (not used):** expanding produces polynomial bounds involving $d(x)$ and $d(y)$, but these are less transparent than rewriting the upper bound in terms of $x, f(y)$, and $d(x) - d(y)$.

- **Linear/translation ansatz (sanity check):** $f(x) = x + c$ works for every $c \geq 0$ by the AM–GM and RMS–AM inequalities; a homogeneous linear trial $f(x) = ax$ is consistent only for $a = 1$.
- **Numerical checks:** `code/sanity_check.py` tests the candidate family on random inputs across twelve orders of magnitude and checks the orbit-alignment estimate. Standalone write-ups of the three proof lemmas are in `lemmas/`.

Current best

The complete characterization is

$$\boxed{f(x) = x + c \quad \text{for every } x > 0, \quad c \geq 0.}$$

The proof first obtains nonnegative, forward-orbit-invariant displacement $d = f - \text{id}$; asymptotic alignment of two arithmetic forward orbits shows that all positive values of d coincide; finally, continuity of the numerical RMS–AM gap and connectedness of $\mathbb{R}_{>0}$ rule out coexistence of displacement 0 with a positive displacement.

Full proof

Put

$$d(t) = f(t) - t \quad (t > 0).$$

1. Forward orbits and nonnegative displacement

(Standalone version: [orbit-structure lemma](#).) Fix $y > 0$ and substitute $x = f(y)$. The two outer expressions in the given chain are then both $f(y)$, since

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Consequently equality holds throughout, and hence

$$f(f(y)) = 2f(y) - y. \quad (1)$$

It follows that

$$d(f(y)) = f(f(y)) - f(y) = f(y) - y = d(y). \quad (2)$$

For $n \geq 0$, let f^n denote the n -fold iterate (with f^0 the identity). Equations (1)–(2), followed by induction, give

$$d(f^n(y)) = d(y), \quad f^n(y) = y + nd(y) \quad (n \geq 0). \quad (3)$$

Every $f^n(y)$ is positive. If $d(y) < 0$, however, then the second formula in (3) is negative for every sufficiently large integer n , a contradiction. Therefore

$$d(y) \geq 0 \quad \text{for all } y > 0. \quad (4)$$

2. All positive displacements are equal

(Standalone version: [positive-displacements-equal lemma](#).) Let $a, b > 0$ have positive displacements

$$p = d(a) > 0, \quad q = d(b) > 0.$$

By (3), for all nonnegative integers m, n ,

$$x := f^m(a) = a + mp, \quad y := f^n(b) = b + nq,$$

and $d(x) = p$, $d(y) = q$. Write

$$v = f(y) = b + (n+1)q.$$

The left-hand inequality of the problem (that is, the upper bound on the middle expression) becomes

$$\sqrt{\frac{x^2 + v^2}{2}} \geq \frac{f(x) + y}{2} = \frac{(x+p) + (v-q)}{2}.$$

After multiplication by 2, this yields

$$p - q \leq \Phi(x, v), \quad \Phi(u, v) := \sqrt{2(u^2 + v^2)} - u - v. \quad (5)$$

For every sufficiently large m , choose

$$n = n_m := \left\lfloor \frac{a + mp - (b + q)}{q} \right\rfloor.$$

Then $n_m \geq 0$, and with $x_m = a + mp$ and $v_m = b + (n_m + 1)q$ we have

$$0 \leq x_m - v_m < q. \quad (6)$$

Moreover, $x_m, v_m \rightarrow \infty$. Rationalizing gives, for all positive u, v ,

$$\Phi(u, v) = \frac{(u - v)^2}{\sqrt{2(u^2 + v^2)} + u + v}. \quad (7)$$

Thus (6)–(7) imply

$$0 \leq \Phi(x_m, v_m) \leq \frac{q^2}{x_m + v_m} \rightarrow 0.$$

Applying (5) along this sequence gives $p - q \leq 0$. Interchanging a and b and repeating the same argument gives $q - p \leq 0$. Hence

$$p = q. \quad (8)$$

Therefore, either d vanishes identically, or there is a number $c > 0$ such that

$$d(t) \in \{0, c\} \quad (t > 0). \quad (9)$$

3. The two displacement values cannot coexist

(Standalone version: [no-mixed-displacement lemma](#).) Suppose now that $c > 0$ occurs, and define

$$A = \{t > 0 : d(t) = 0\}, \quad B = \{t > 0 : d(t) = c\}.$$

By (9), these sets are disjoint and cover $\mathbb{R}_{>0}$, and B is nonempty.

If $x \in B$ and $y \in A$, then $f(x) = x + c$ and $f(y) = y$. The left-hand inequality in the problem gives

$$\sqrt{\frac{x^2 + y^2}{2}} \geq \frac{x + c + y}{2},$$

or equivalently

$$c \leq \Phi(x, y), \quad (10)$$

where Φ is as in (5). The function Φ is continuous on $\mathbb{R}_{>0}^2$ and satisfies

$$\Phi(t, t) = 0 \quad (t > 0). \quad (11)$$

Fix $b \in B$. By (11) and continuity, all y sufficiently close to b satisfy $\Phi(b, y) < c$. Such a y cannot belong to A , by (10), so it belongs to B . Hence B is open in $\mathbb{R}_{>0}$. Similarly, if $a \in A$, then all x sufficiently close to a satisfy $\Phi(x, a) < c$; (10) shows that such an x cannot belong to B . Hence A is also open in $\mathbb{R}_{>0}$.

The interval $\mathbb{R}_{>0}$ is connected, so it cannot be the union of two disjoint nonempty open subsets. Since B is nonempty, we conclude that A is empty. Thus $d(t) = c$ for every $t > 0$. Together with the case $d \equiv 0$, this proves that every solution must have the form

$$f(t) = t + c \quad (t > 0)$$

for some $c \geq 0$.

4. Verification

Conversely, fix $c \geq 0$ and let $f(t) = t + c$. For all $x, y > 0$, set $z = y + c > 0$. Then

$$\frac{f(x) + y}{2} = \frac{x + z}{2}.$$

The RMS–AM and AM–GM inequalities applied to the positive numbers x, z give

$$\sqrt{\frac{x^2 + z^2}{2}} \geq \frac{x + z}{2} \geq \sqrt{xz}.$$

Since $z = f(y)$, this is precisely the required chain. Hence every $c \geq 0$ indeed gives a solution, completing the characterization.